

Juhwan Lee

Phone: (+82)10-2254-8108 | Email: kokomoty97@gmail.com

SUMMARY

Doctoral student and research assistant in Air Transportation at Korea Aerospace University, with experience in airport operations, capacity and delay analysis, passenger terminal simulation, and data-driven prediction. My research focuses on extending airport simulation and Transformer-based queue forecasting toward AI-enabled decision support for aviation systems.

EDUCATION

Korea Aerospace University (KAU), Goyang, South Korea *Sep 2023 – Present*

Ph.D. Student in Air Transportation, Transportation and Logistics

Research focus: airport operations, capacity analysis, passenger terminal simulation, and data-driven queue forecasting

Korea Aerospace University (KAU), Goyang, South Korea *August 2023*

M.S. in Air Transportation, Transportation and Logistics

GPA: 4.5/4.5

Related courses: Airport Design with Simulation, Theory and Applications of Data Mining

Korea Aerospace University (KAU), Goyang, South Korea *August 2021*

B.S. in Air Transportation, Transportation and Logistics

GPA: 4.05/4.5

Related courses: Introductory Applied Machine Learning, Air Traffic Systems Analysis, Airspace Systems Analysis, Computer Applications in Air Traffic Management

RESEARCH INTERESTS

Data-Driven Airport Operations and Aviation Decision Support

- Airport capacity and demand–capacity imbalance analysis
- Passenger terminal simulation and data-driven congestion forecasting
- AI-enabled decision support for airport operations and resource allocation

PUBLICATIONS

- [1] **Juhwan Lee**, S. Yoon, K. Lee, S. Jeong, and H. Baik, “Data-Driven Airport Passenger Queue Forecasting for Departure Gates and Security Checkpoints,” IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), 2026.
- [2] J. Shin, H. Baik, **Juhwan Lee**, and B. Yi, “Network Connectivity Analysis of Major Asian Airports Using the NetScan Model:Focusing on Incheon International Airport,” Journal of the Korean Society for Aviation and Aeronautics, 2025.
- [3] J. Shin, B. Yi, M. Kim, J. Cho, and **Juhwan Lee**, “Study on Airport Capacity Utilization Analysis Using the Capacity Coverage Chart (CCC),” Journal of the Korean Society for Aviation and Aeronautics, 2025.
- [4] E. Kim, **Juhwan Lee**, Y. Shin, and H. Baik, “A Study on Airport Access Mode Choice Behavior Using Mobile Data: Case of Incheon International Airport,” IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), 2023.
- [5] Y. Lee, **Juhwan Lee**, and K. Lee, “A Study on the Visualization of an Airline’s Fleet State Variation,” *Journal of the Korean Society for Aviation and Aeronautics*, 2021.

PROJECT EXPERIENCE

Development of Passenger Terminal Simulation Models for Incheon International Airport (Jul 2025 - Dec 2025)

- Developed MATLAB-based simulation models for departure passenger processing facilities using actual airport observation data.
- Estimated queue length and waiting time under time-varying demand and operational conditions.

Strategic Planning for Enhancing Hub Competitiveness of Incheon International Airport (Oct 2025 - Dec 2025) [2],[3]

- Analyzed changes in delay according to passenger demand and capacity conditions.
- Supported strategic evaluation of airport competitiveness from an operational performance perspective.

Analysis of Variable Multiple Taxi-Line Operations for Apron Congestion Mitigation at Jeju International Airport (Jul 2025 - Dec 2025)

- Simulated apron congestion and aircraft surface movement using TAAM.
- Evaluated taxi-line operational alternatives for congestion mitigation and surface movement efficiency.

Master Plan for Incheon International Airport (Sep 2022 - Aug 2024) [4]

- Performed runway capacity analysis using TAAM and passenger terminal simulation using CAST.
- Evaluated capacity, congestion, delay, and economic feasibility for airport master planning.

Research on Establishing Airport Capacity Setting Guidelines (Sep 2021 - Dec 2021)

- Supported the development of airport capacity setting guidelines for airport planning and management.
- Reviewed capacity concepts, operational constraints, and demand–capacity relationships.

AWARDS AND ACHIEVEMENTS

Scholarship, KAU Professional Talent Development Scholarship	\$4,520	<i>Fall 2016</i>
2nd Place in Capstone Design Competition		<i>Fall 2020</i>
2nd Place in ICT Startup Makerthon		<i>Fall 2019</i>
Scholarship, Jeongseok C	\$841	<i>Fall 2019</i>
Scholarship, Jeongseok C	\$739	<i>Fall 2016</i>

TECHNICAL SKILLS

Programming	Python, MATLAB, $\text{\LaTeX}$
AI / Data Analysis	PyTorch, Time-series forecasting, Transformer-based modeling, Statistical analysis
Simulation	TAAM for runway and surface operation simulation; CAST and MATLAB-based passenger terminal simulation
Aviation Analysis	Airport capacity and delay analysis, Queue and waiting time analysis